

CSCI-246 Discrete Structures HW 06

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Objective

- Understanding sum and product rules.
- Understanding probability and random variables.
- Understanding the problem solving process.
- Understanding counting.
- Understanding asymptotic analysis.
- Understanding properties of Big O.
- Understanding algorithm analysis.
- Understanding problem solving process.

Submission requirements

- *Type or clearly hand-write* your solutions into a **PDF FORMAT**.
- **DO NOT UPLOAD images.**
- *non-pdf or emailed solutions will not be graded.*
- If you take pictures of your handwritten homework, put it into pdf format.
- *Start each problem in a new page.*

- Follow the model that you have learned during the lectures for proofs.
- Do not wait until the last minute to submit the assignment.
- You can submit any number of times before the deadline.
- If you are using latex, and you do not know how to type a symbol, use the following website. You can draw the symbol here and it will give you the latex code and the packages that you have to import. <https://detexify.kirelabs.org/classify.html>
- If you are using latex to write the answer, you can use overleaf to make your life easier. **Overleaf is a free, online platform that helps users create and publish scientific and technical documents using LaTeX, a markup-based document preparation system**
- If you do not understand a problem, ask questions during/after the lectures, or during office hours or via discord.
- ***Do not use generative AI to write answers.***

1 Q1

Suppose Inclusion-Exclusion for three sets A, B, C is as following:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

1. Let $A = \{0, 1, 2, 3, 4\}$, $B = \{0, 2, 4, 6\}$, $C = \{0, 3, 6\}$. Find the value of $A \cup B \cup C$ using inclusion-exclusion rule. **(You have to use the above rule).**
2. Consider the words **ONE, TWO, THREE, FOUR, FIVE, SIX, SEVEN, and EIGHT**. Let E be the set of these words containing at least one **E**, let T be the words containing a **T**, and let R be the words containing an **R**. Then, calculate $|E \cap T \cap R|$.

2 Q2

In the United States, a text message can be sent either to a regular 10-digit phone number, or to a so-called short code which is a 5- or 6-digit number. Neither a phone number nor a short code can start with a 0 or a 1. How many different textable numbers are there in the United States?.

Hint: Use the product rule.

3 Q3

Sam flips a fair coin 100 times. Let the outcome be the number of heads that he sees.

1. What is the sample space? (represent the sample space using sets).
2. For Sam's 100 flips, what is the $Pr[0]$ (What is the probability that you see 0 heads)?
3. For Sam's 100 flips, what is $Pr[50]$?
4. For Sam's 100 flips, what is $Pr[64]$?

4 Q4

If we roll a fair 3-sided die 11 times, what is the number of ways that we can get 4 1's, 5 2's, and 2 3's? Show your work.

5 Q5

1. Prove that $3n^4 + 4n^2 - 2n = O(n^4)$ by constructing $c > 0, n_0 \geq 0 : \forall n \geq n_0 : 3n^4 + 4n^2 - 2n \leq c \cdot n^4$.

6 Q6

In class we proved that $n^3 \neq O(n^2)$. Following that proof, fill the blanks, to show that $3n^4 + 4n^2 - 2n \neq O(n^3)$ by disproving $\exists c > 0, n_0 \geq 0 : \forall n \geq n_0 : 3n^4 + 4n^2 - 2n \leq c \cdot n^3$.

To disprove $\exists c > 0, n_0 \geq 0 : \forall n \geq n_0 : 3n^4 + 4n^2 - 2n \leq c \cdot n^3$, we need to show that:

So we show how to construct -----, given any -----, and -----.

Let $c > 0$ and $n_0 \geq 0$. Let $n = \max(n_0, c + 1)$, then consider $n = c + 1$.

$3n^4 + 4n^2 - 2n =$ fill in here and take as much as lines you need

which is $3n^4 + 4n^2 - 2n > c \cdot (c + 1)^3$. Therefore, we have shown how to produce a $n \geq n_0$ such that $3n^4 + 4n^2 - 2n > c \cdot n^3$, for any c, n_0 value, meaning that $3n^4 + 4n^2 - 2n$ is not $O(n^3)$.

7 Q7

1. Give two functions, f and h , such that $f = O(n^3)$ and $h = O(2^n)$ but $f(n) \neq O(h(n))$.

Note that you only need to pick f, h that follows the given restriction.

2. Now that you have picked f, h , prove that $f = O(n^3)$ and $h = O(2^n)$.
3. Use that same f, h functions and prove that $f(n) \neq O(h(n))$.